# Abhimanyu Susobhanan

## Areas of Interest

Gravitational Waves, Pulsars, Data Analysis Methods, Astrophysical Software

## Education

Aug 2015– Sep 2021 **Master of Science in Physics + Doctor of Philosophy in Astrophysics**, *Department of Astronomy & Astrophysics, Tata Institute of Fundamental Research, Mumbai, Maharashtra, India*
***Thesis Title***: Perspectives in nanohertz gravitational-wave astronomy
***Advisor***: Prof. Achamveedu Gopakumar

Aug 2008– May 2012 **Bachelor of Technology in Physical Sciences**, *Department of Earth & Space System Sciences, Indian Institute of Space Science and Technology, Thiruvananthapuram, Kerala, India*
***CGPA***: 8.27/10

## Career

Nov 2025 – *present* **Assistant Professor**, *Indian Institute of Science Education and Research Thiruvananthapuram, Thiruvananthapuram, Kerala, India*

Apr 2024 – Oct 2025 **Postdoctoral Fellow**, *Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Hannover, Lower Saxony, Germany*

Jun 2022 – Feb 2024 **Postdoctoral Fellow**, *Center for Gravitation Cosmology and Astrophysics, University of Wisconsin-Milwaukee, Milwaukee, Wisconsin, USA*

Sep 2021 – Jan 2022 **Postdoctoral Fellow**, *National Centre for Radio Astrophysics, Tata Institute of Fundamental Research, Pune, Maharashtra, India*

Aug 2015– Sep 2021 **Research Scholar**, *Department of Astronomy & Astrophysics, Tata Institute of Fundamental Research, Mumbai, Maharashtra, India*

Jul 2019– Oct 2019 **Visiting Scholar**, *CSIRO Astronomy and Astrophysics, Marsfield, NSW, Australia*

Sep 2012– Jun 2015 **Scientist/Engineer**, *Liquid Propulsion Systems Centre, Indian Space Research Organization, Valiyamala, Thiruvananthapuram, Kerala, India*

Jan 2012– Apr 2012 **Intern**, *U.R. Rao Satellite Centre, Indian Space Research Organization, Bengaluru, Karnataka, India*

Jun 2011– Aug 2011 **Intern**, *Raman Research Institute, Bengaluru, Karnataka, India*

Jan 2010– Feb 2010 **Intern**, *Physical Research Laboratory, Ahmedabad, Gujarat, India*

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## Selected Publications

2026* [1] Vidit Singh, Achamveedu Gopakumar, **Abhimanyu Susobhanan**, and Subhajit Dandapat, "*Incorporating post-Newtonian accurate gravitational wave emission effects in pulsar timing array signals from individual supermassive black hole binaries*", Submitted to Physical Review D

2026* [2] Churchil Dwivedi and **Abhimanyu Susobhanan**, "*PSRDISP: A Novel Approach to Modeling Dispersive Processes in Pulsar Timing*", Submitted to Journal of High Energy Astrophysics, arXiv: 2607.12609

2026* [3] Massimo Vaglio, Amodio Carleo, **Abhimanyu Susobhanan**, et al., "*Constraints on Einstein-aether gravity from the precision timing of PSR J1738+0333*", Submitted to Physical Review D, arXiv: 2605.01436

2026 [4] Gabriella Agazie, David Kaplan, **Abhimanyu Susobhanan**, et al., "*CHIME-o-Grav: Wideband Timing of Four Millisecond Pulsars from the NANOGrav 15-yr dataset*", The Astrophysical Journal, 1002, 2, DOI: 10.3847/1538-4357/ae563f

2026 [5] **Abhimanyu Susobhanan**, Avinash Kumar Paladi, Réka Desmecht, et al., "*Revisiting wideband pulsar timing measurements*", Monthly Notices of the Royal Astronomical Society, 547, stag444, DOI: 10.1093/mnras/stag444

2026 [6] K Nobleson, ..., **Abhimanyu Susobhanan**, et al., "*The Indian Pulsar Timing Array Data Release 2: II. Customised Single-Pulsar Noise Analysis and Noise Budget*", Journal of High Energy Astrophysics, 53, 100594, DOI: 10.1016/j.jheap.2026.100594

2025 [7] **Abhimanyu Susobhanan**, "*Bayesian pulsar timing and noise analysis with Vela.jl: the wideband paradigm*", Physical Review D, 112, 063023, DOI: 10.1103/n3ck-lfdy

2025 [8] **Abhimanyu Susobhanan** and Rutger van Haasteren, "*Gaussian process representation of dispersion measure noise in pulsar wideband data sets*", Monthly Notices of the Royal Astronomical Society, 542, 2892–2900, DOI: 10.1093/mnras/staf1422

2025 [9] **Abhimanyu Susobhanan**, "*Bayesian pulsar timing and noise analysis with Vela.jl: an overview*", The Astrophysical Journal, 980, 165, DOI: 10.3847/1538-4357/adaaec

2024 [10] **Abhimanyu Susobhanan**, David Kaplan, Anne Archibald, et al., "*PINT: Maximum-likelihood estimation of pulsar timing noise parameters*", The Astrophysical Journal, 971, 150, DOI: 10.3847/1538-4357/ad59f7

2024 [11] Subhajit Dandapat, **Abhimanyu Susobhanan**, et al., "*Efficient prescription to search for linear gravitational wave memory from hyperbolic black hole encounters and its application to the NANOGrav 12.5-year dataset*", Physical Review D, 109, 103018, DOI: 10.1103/PhysRevD.109.103018

2024 [12] Gabriella Agazie, ..., **Abhimanyu Susobhanan**, et al., "*The NANOGrav 12.5-year data set: A computationally efficient eccentric binary search pipeline and constraints on an eccentric supermassive binary candidate in 3C 66B*", The Astrophysical Journal, 963, 144, DOI: 10.3847/1538-4357/ad1f61

2023 [13] Avinash Kumar Paladi, ..., **Abhimanyu Susobhanan**, et al., "*Multi-band Extension of the Wideband Timing Technique*", Monthly Notices of the Royal Astronomical Society, 527, 213–231, DOI: 10.1093/mnras/stad3122

2023 [14] Aman Srivastava, ..., **Abhimanyu Susobhanan**, et al., "*Noise analysis in the Indian Pulsar Timing Array Data Release I*", Physical Review D, 108, 023008, DOI: 10.1103/PhysRevD.108.023008

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2023 [15] **Abhimanyu Susobhanan**, "*Post-Newtonian-accurate pulsar timing array signals induced by inspiralling eccentric binaries: accuracy, computational cost, and single-pulsar search*", Classical and Quantum Gravity, 40, 155014, DOI: 10.1088/1361-6382/ace234

2023 [16] Subhajit Dandapat, Michael Ebersold, **Abhimanyu Susobhanan**, et al., "*Gravitational Waves from Black-Hole Encounters: Prospects for Ground- and Galaxy-Based Observatories*", Physical Review D, 108, 024013, DOI: 10.1103/PhysRevD.108.024013

2022 [17] Pratik Tarafdar, ..., **Abhimanyu Susobhanan**, et al., "*The Indian Pulsar Timing Array: First data release*", Publications of the Astronomical Society of Australia, 39, E053, DOI: 10.1017/pasa.2022.46

2022 [18] K Nobleson, ..., **Abhimanyu Susobhanan**, et al., "*Low-frequency wideband timing of InPTA pulsars observed with the uGMRT*", Monthly Notices of the Royal Astronomical Society, 512, 1234-1243, DOI: 10.1093/mnras/stac532

2021 [19] Jaikhomba Singha, ..., **Abhimanyu Susobhanan**, et al., "*Evidence for profile changes in PSR J1713+0747 using the uGMRT*", Monthly Notices of the Royal Astronomical Society: Letters, 507, L57–L61, DOI: 10.1093/mnrasl/slab098

2021 [20] Lankeswar Dey, ..., **Abhimanyu Susobhanan**, et al., "*Explaining temporal variations in the jet position angle of blazar OJ 287 using its binary black hole central engine model*", Monthly Notices of the Royal Astronomical Society, 503, 3, 4400–4412, DOI: 10.1093/mnras/stab730

2021 [21] **Abhimanyu Susobhanan**, Yogesh Maan, Bhal Chanda Joshi, et al., "*pinta: The uGMRT Data Processing Pipeline for the Indian Pulsar Timing Array*", Publications of the Astronomical Society of Australia, 38, E017, DOI: 10.1017/pasa.2021.12

2020 [22] **Abhimanyu Susobhanan**, Achamveedu Gopakumar, George Hobbs, and Stephen Taylor, "*Pulsar timing array signals induced by black hole binaries in relativistic eccentric orbits*", Physical Review D, 101, 043022, DOI: 10.1103/PhysRevD.101.043022

2018 [23] **Abhimanyu Susobhanan**, Achamveedu Gopakumar, Bhal Chanda Joshi, and Ranjan Kumar, "*Exploring the effect of periastron advance in small-eccentricity binary pulsars*", Monthly Notices of the Royal Astronomical Society, 480, 5260-5271, DOI: 10.1093/mnras/sty2177

2017 [24] Yannick Boetzel, **Abhimanyu Susobhanan**, Achamveedu Gopakumar, Antoine Klein, and Philippe Jetzer, "*Solving post-Newtonian accurate Kepler equation*", Physical Review D, 96, 044011, DOI: 10.1103/PhysRevD.96.044011

## Teaching

Spring 2026 **Instructor**, *Indian Institute of Science Education and Research Thiruvananthapuram*
Course Title: Electrodynamics

Autumn 2023 **Substitute Instructor**, *University of Wisconsin-Milwaukee*
Course Title: Survey of Astronomy (Instructor: Sarah Vigeland)

Autumn 2018 **Teaching Assistant**, *Tata Institute of Fundamental Research*
Course Title: Astronomy and Astrophysics I (Instructors: H. M. Antia & A. Gopakumar)

Autumn 2017 **Teaching Assistant**, *Tata Institute of Fundamental Research, Mumbai, India*
Course Title : Electrodynamics II (Instructors: Sushil Mujumdar & A. Gopakumar)

Spring 2016 **Teaching Assistant**, *Tata Institute of Fundamental Research, Mumbai, India*
Course Title: Astronomy and Astrophysics II (Instructor: Manoj Puravankara)

## Skills

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| Programming Languages | C++, Python, C, Julia, Wolfram Language, LaTeX, bash |
| Astrophysical software | PINT, TEMPO2, ENTERPRISE, PSRCHIVE, DSPSR |
| Telescope observations | Giant Metre-wave Radio Telescope, Parkes Radio Telescope |
| Data analysis | Bayesian inference, Data visualization |

## Languages

Malayalam (native), English, Hindi, German (beginner)

## Awards and Fellowships

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| 2019 | **Ratanbai Jerajani Award** for the best seminar in the area of Astronomy and Astrophysics at TIFR |
| 2019 | **Sarojini Damodaran Fellowship** for international travel |
| 2006-2012 | **National Talent Search Scholarship** |

## References

**Dr. Rutger van Haasteren**
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Hannover, Lower Saxony, Germany
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**Prof. David Kaplan**
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